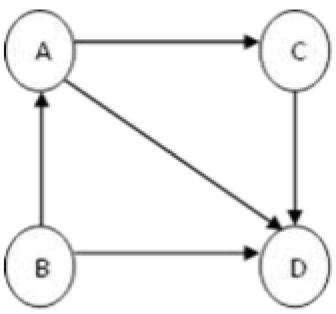
DATA STRUCTURES - 2024

SECTION – A

1. Define array.
2. What is meant by Recursive function? Give example.
3. Compare Structure and Union.
4. What is the use of preprocessor directives in C?
5. What is meant by an Abstract Data Type (ADT)?
6. List the applications of stack?
7. What is meant by Binary Search Tree?
8. What is meant by “degree of a graph”?
9. Define sorting.
10. What are drawbacks of bubble sort?

SECTION – B

11. A. Explain the function (i) strlen() (ii) strcpy (iii) strcat() (iv) strcmp() with example.
    B. What do you mean by call by reference and call by value. Write a program in c to exchange the values of two variables.
12. A. Define a structure called student that would contain name, regno and marks of five subjects and percentage. Write a C program to read the details of name, regno and marks of five subjects for 30 students and calculate the percentage and display the name, regno, marks of 30 subjects and percentage of each student.
    B. Explain about the fopen (), fclose(), feof(), fprintf(), fscanf() and fseek()?
13. A. Explain the following operations performed on the doubly linked list:
    i. Inserting an element in beginning of the list.
    ii. Deleting an element in the end of the list.
    iii. Display all the elements in the list.
    B. What is Stack? Explain various operations of Stack using linked list.
14. A. What are expression tree? Write the procedures for constructing an expression tree with an example.
    B. Explain Depth First Search on a graph with necessary data structures.

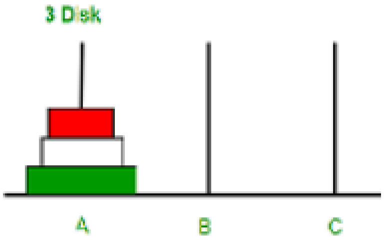


15. A. Describe the Quick sort algorithm by sorting the following collection of numbers as an example, 42, 7, 52, 57, 62, 37, 32, 27, 22.
    B. Compare linear search and binary search. State and explain the algorithm for both the search with example

SECTION – C

16. A. You are given a set of three pegs and n disks, with each disk a different size. Let's name the pegs A, B, and C, and let's number the disks from 1, the smallest disk to n, the largest disk. At the outset, all n disks are on peg A, move the plates from A to C with following rules.

    1. You may move only one disk at a time.
    2. No disk may ever rest on top of the smaller disk.
    3. A 3 rd peg could be used as an intermediate to store one or more disk while they were been moved from source to destination.



B. Illustrate the intermediate steps of heap sort for the given set of numbers 67, 3, 24, 45, 9, 12, 30, 29, 90